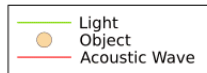


Unsupervised learning for the variable sound speed and inverse operator in photoacoustic tomography

Dabin Park
(working with G. Hwang, G. Jeon and S. Moon)

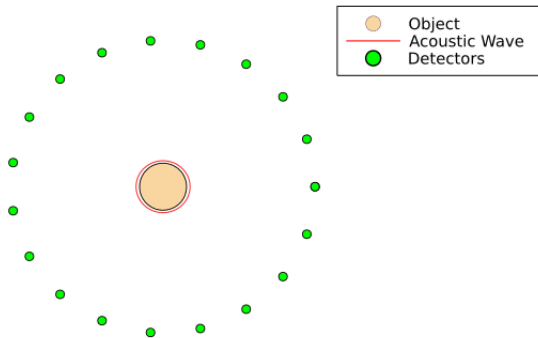
Kyungpook National University

KDISS
Nov. 3, 2023



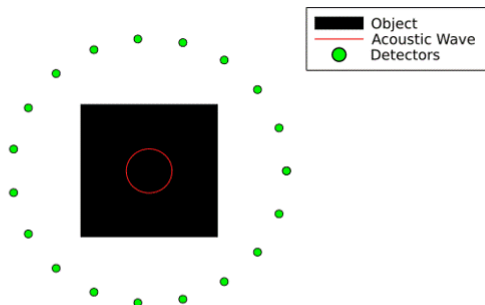
PhotoAcoustic Effect

출처 : <https://freshrimpsushi.github.io/posts/principles-of-photocoustic-tomography/>



PhotoAcoustic Tomography

출처 : <https://freshrimpsushi.github.io/posts/principles-of-photocoustic-tomography/>



PhotoAcoustic Tomography

출처 : <https://freshrimpsushi.github.io/posts/principles-of-photocoustic-tomography/>

$$\begin{cases} \partial_t^2 p(\mathbf{x}, t) = c(\mathbf{x}) \Delta_{\mathbf{x}} p(\mathbf{x}, t) & (\mathbf{x}, t) \in \mathbb{R}^2 \times [0, \infty) \\ p(\mathbf{x}, 0) = f(\mathbf{x}) & \mathbf{x} \in \mathbb{R}^2 \\ \partial_t p(\mathbf{x}, t)|_{t=0} = 0 & \mathbf{x} \in \mathbb{R}^2 \end{cases}$$

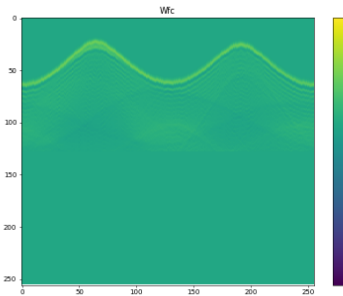
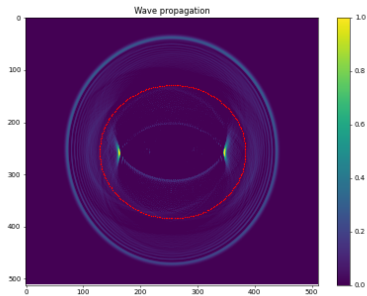
- $c(\mathbf{x})$ is the sound speed at $\mathbf{x}$
- f is the initial pressure.
(initial pressure contains biological information, such as the location of cancer cells in a physically small tissue sample.)

$\Rightarrow$ • f has compact support within the boundary region Ω .

- detectors are located at the boundary of the region $\partial\Omega$.

$$\mathcal{W} : (f, c) \mapsto p(\mathbf{x}, t)|_{(\mathbf{x}, t) \in \partial\Omega \times [0, \infty)} = \mathcal{W}(f, c)$$

$$f \text{ or } c \leftarrow \mathcal{W}(f, c)$$



- **Supervised Learning**

A type of machine learning where a model is trained on labeled data.

e.g.) Image Classification, Regression problem

- **Unsupervised Learning**

A type of machine learning that deals with identifying patterns in data.

Unsupervised learning is that the data used to train the model is not labeled

e.g.) Clustering, Anomaly Detection

- **Self-supervised Learning**

Learning from unlabeled data and can be seen as an intermediate form between supervised and unsupervised learning.

e.g.) Data Generation, Feature Extraction

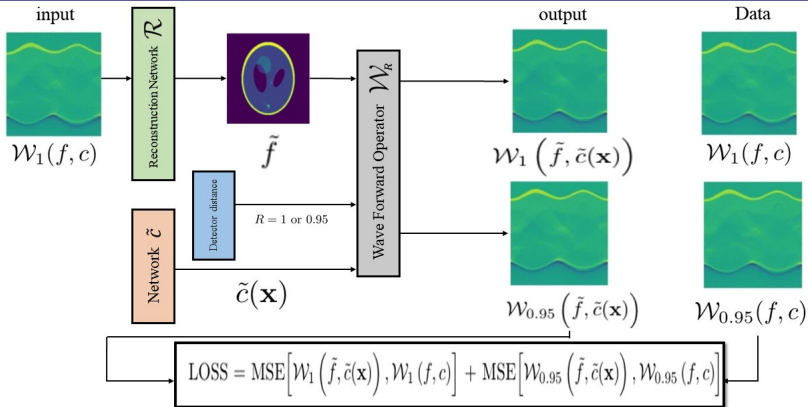
Hypothesis 1

If the collection of the Dirichlet and Neumann data pair $\{(p_f^c(\mathbf{x}, t)|_{(\mathbf{x}, t) \in \partial\Omega \times [0, \infty)}, \partial_\nu p_f^c(\mathbf{x}, t)|_{(\mathbf{x}, t) \in \partial\Omega \times [0, \infty)}) : f \in L^2\}$ is given,

then c and the inverse operator is uniquely determined.

Here ν denotes the outward normal to $\partial\Omega$ and so $\partial_\nu = \nu \cdot \nabla$ is normal derivative.

Network Design

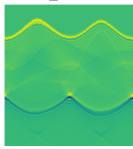


MSE : Mean Squared Error

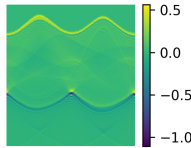
Let $\mathbf{x} = \{x_i\}_{i=1}^N$, $\mathbf{y} = \{y_i\}_{i=1}^N$.

Train in epoch=66, index=989

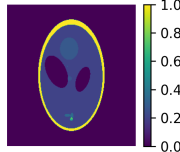
input_data Wfc



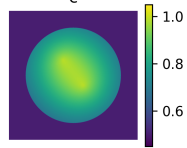
data NWfc



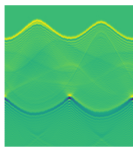
f



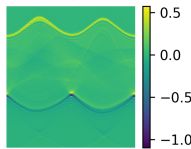
c



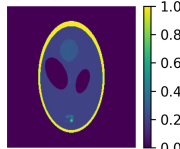
Wfc_avg=0.06962,
Wfc[989]=0.06953



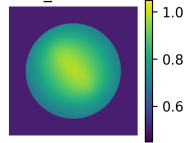
NWfc_avg=0.07954,
NWfc[989]=0.07876



f_avg=0.06817,
f[989]=0.05319



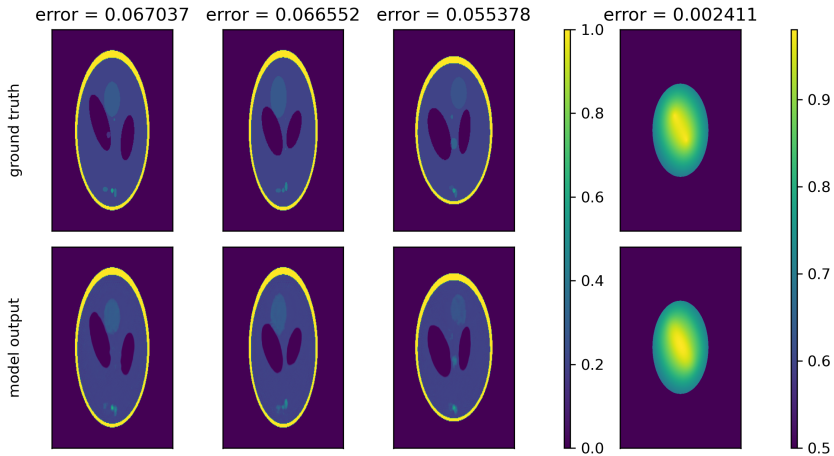
error_c=0.00331



$$\text{error for } f = \frac{1}{N} \sum_{i=1}^N \frac{\|f_i^{\text{gt}} - f_i\|_2}{\|f_i^{\text{gt}}\|_2},$$

where f^{gt} is ground truth for f

$$\|c^{\text{gt}} - c\|_2$$



Reference



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Photoacoustic Tomography.

CRC Press, 2018.



A. G. Bell.

On the production and reproduction of sound by light.

Proceedings of American Association for the Advancement of Science, volume 29, pages 115–136, 1881.



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On the uniqueness and unsupervised learning for the variable sound speed and inverse operator in photoacoustic tomography

Thank you.